

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Ravi KapoorStudent ID: 002253839**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H)P(H) / P(K=3)$$

$$P(K=3|H) = 4/16 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4/16(.25) + .25(.75)$$

$$= .029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .029 / .2904$$

$$= 0.354$$

see back →

cont. from front

$$= .2904$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$i = .04 = .670$$

$$= .4116(.25) + .25(.75)$$

$$8/5.5 = 1.4545$$

$$3.5/9 = .389$$

$$P(H|K=3) = .1029 / .2904$$

$$3.84 \times .4 = 1.536$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$1/2.49 = .401$$

$$= 0.354$$

x2?

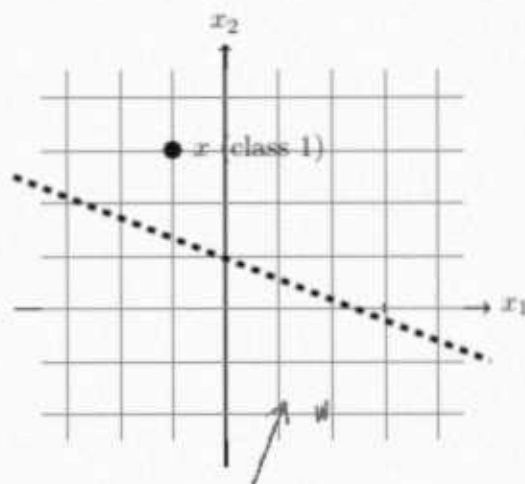
units?

Name: Ravi Kapor

Student ID: 002253839

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x = [-1, 3]: -0.39 + 3 = 0.94$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

Name: Ravi KapoorStudent ID: 002253839**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$p = \frac{1}{1 + e^{-z}}$$

$$P(\text{pass} | x=4) = 0.62$$

$$e^{-0.4} = 0.670$$

$$\ln 0.5 = -0.693$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -5, 0, 1, 1.5$$

$$p = 0.269, 0.378, 0.5, 0.731, 0.818$$

$$L = -[\ln(0.731) + \ln(0.622) + \ln(0.5) + \ln(0.731) + \ln(0.818)]$$

$$= 313 + 475 + 693 + 313 + 201$$

$$L = 200$$

ans

Name: Ravi KapoorStudent ID: 002253839**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

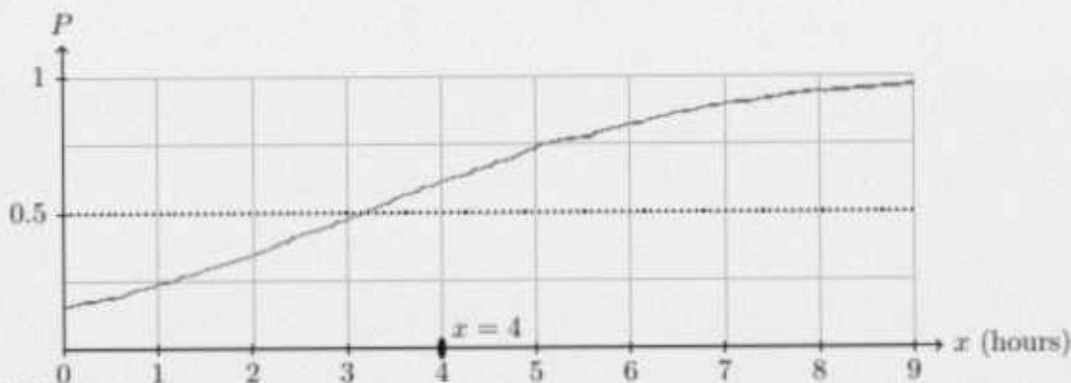
$$\begin{aligned} dL/dw &= .269 + .378(2) - .5(3) - .269(5) - .182(6) \\ &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \end{aligned}$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = 5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

- (d) Sketch $P(\text{pass} \mid x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$x = 1.5 / .5 = 3 \text{ hours}$$

$$b \times 3.75 = 225$$

Q3 cont

$$\sqrt{2.49} = .401$$

$$384 \times .4 = .1536$$

$$= 269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$\approx 2.203$$

$$b \times .375 = 225$$

x27

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Name: Elena PetrovaStudent ID: 002428429**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

$$= .3786$$

$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

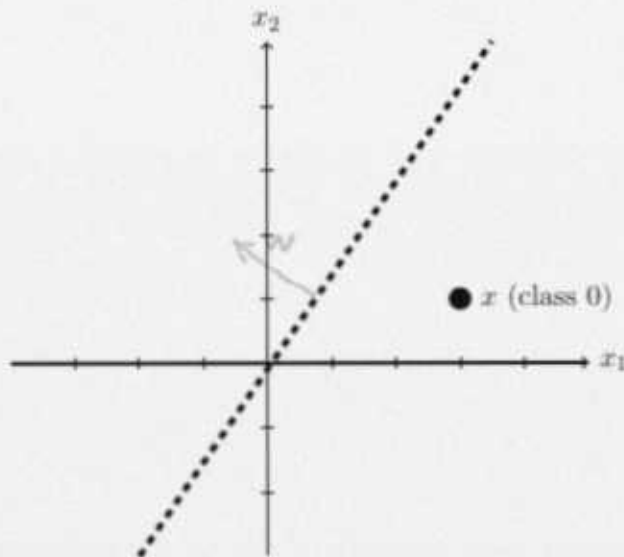
$$8/5.5 = 1.4545$$

$$\ln .5 = -.693$$

$$.6 \times .375 = .225$$

Name: Elena PetrovaStudent ID: 002428429**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

~~slope = 5.5/8~~

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45 (x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$\rightarrow$ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Elena PetrovaStudent ID: 002428429**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

hmm

Name: Elena PetrovaStudent ID: 002428429

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

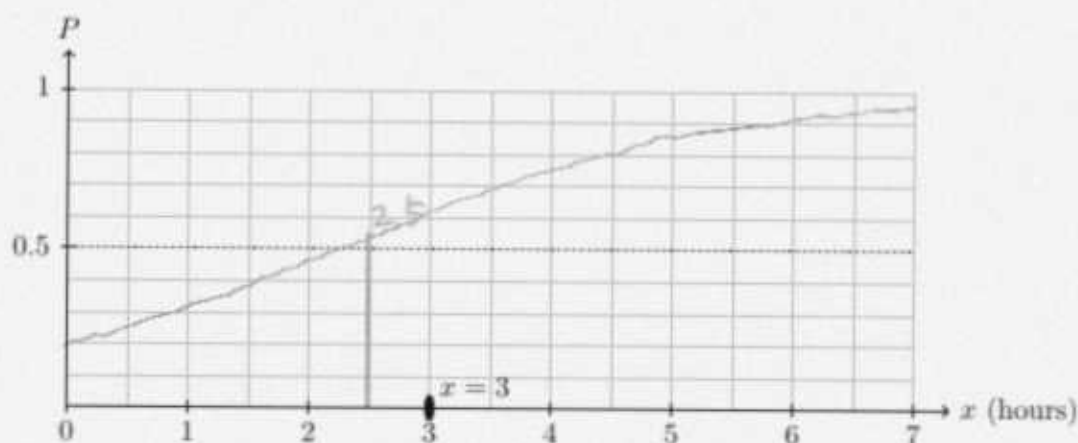
$$w = 8 - 1(-.486) = 8.49$$

$$b = -2 - 1(.282) = -2.28$$

$$8/5.5 = 1.4545$$

$$1/2.49 = .401$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



Switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

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Name: Asha MeenonStudent ID: 008612993**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .3786$$

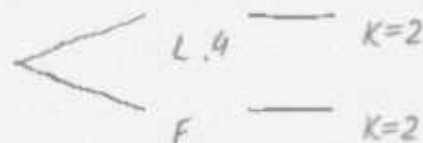
$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

$$.384 \times .4 = .1536$$

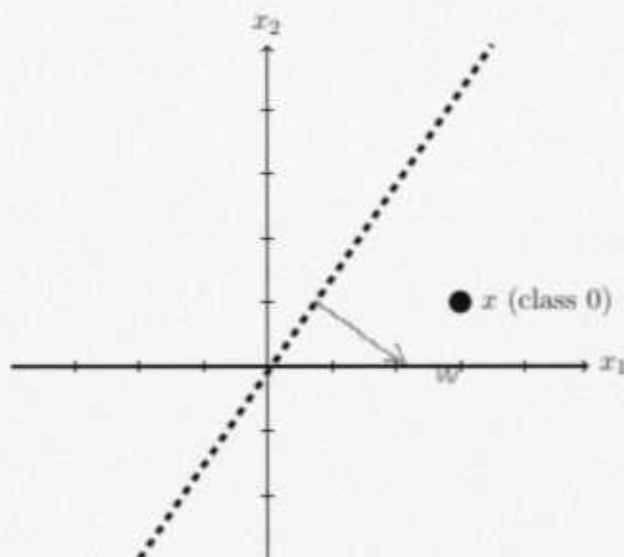
$$1/2.49 = .401$$

$$2 \times .378 = .756$$



Name: Asya MenonStudent ID: 008612993**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2)) \\ = 8 / 5.5 \approx 1.45$$

$$x_2 = 1.45 x_1 - 0.09 \\ w = [1, 3.5]$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$\rightarrow$ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

$$2 \times .378 = .756$$

see back $\rightarrow$

$x_2?$

cont. from front

$$1 + 1.49 = 2.49$$

$$m_S = -.893$$

$$= 8 / 5.5 = 1.45$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

(b) no, any $c > 0$ scales (w, b)

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$8 / 5.5 = 1.4545$$

1029/2904

Name: Asha MenonStudent ID: 008612995**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\text{sigma}(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

$$1/2.49 = .401$$

$$3.5/9 = .389$$

$$\ln .5 = -.693$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L = .231 + .401 + .769$$

$$L \sim 1.12$$

Name: Ashu MenonStudent ID: 008612993**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$x = 0.5$$

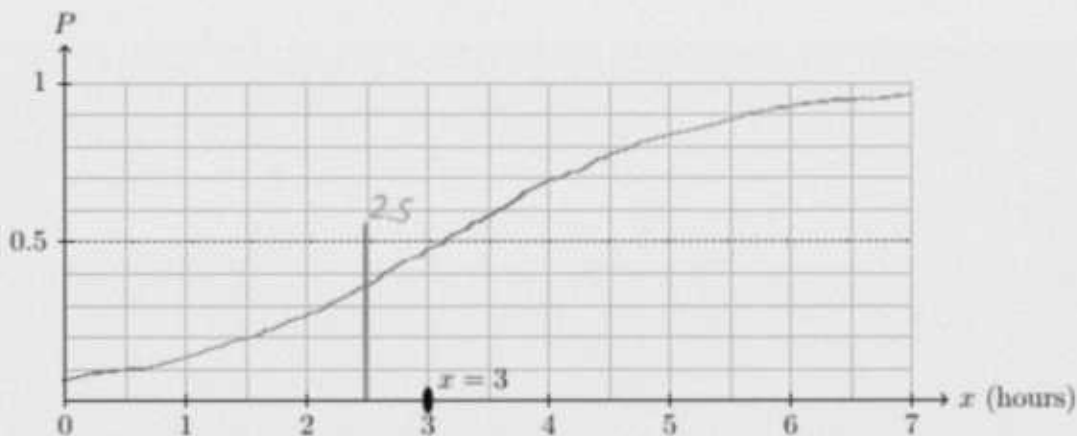
$$\begin{aligned} dL/dw &= .231(1) + .401(2) - .231(4) - .119(5) \\ &= .231 + .802 - .924 - .595 = -.486 \end{aligned}$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

$$\hat{e} - 0.4 = .670$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$8/5.5 = 1.4545$$

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Name: Hana NovakStudent ID: 007394770**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(L \mid K=2) &= P(K=2 \mid L) P(L) / P(K=2) \\
 P(K=2 \mid L) &= .384 \quad P(K=2 \mid F) = .375 \\
 P(K=2) &= P(K=2 \mid L) P(L) + P(K=2 \mid F) P(F) \\
 &= .384(.4) + .375(.6) \\
 &= .1536 + .225
 \end{aligned}$$

L .4 — K=2
F — K=2

$$= .3786$$

$$P(L \mid K=2) = .1536 / .3786$$

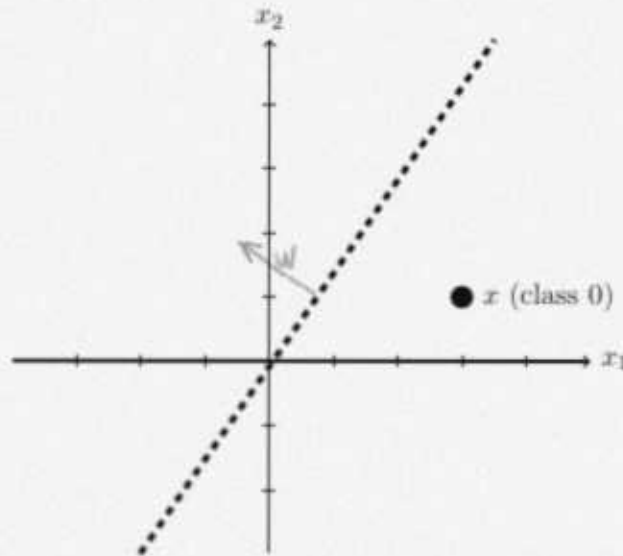
$$\ln .5 = -.693$$

$$e^{-0.4} = .670$$

$$.1029 / .2904$$

Name: Hana NovakStudent ID: 007394770**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\begin{aligned}
 &= 8 / 5.5 \sim 1.45 \\
 &x_2 + 3 = 1.45 (x_1 + 2) \\
 &x_2 = 1.45 x_1 - 0.09 \\
 &w = [1.45, -1], b = -0.09 \\
 &x = [3, 1]: 4.35 - 1 = 3.35 > 0 \\
 &\rightarrow \text{class 1, wrong side! flip} \\
 &(b) \text{ no, any } c > 0 \text{ scales } (w, b) \\
 &w = [-1.45, 1], b = 0.09 \\
 &3.5/9 = .389 \\
 &1 + 1.49 = 2.49
 \end{aligned}$$

Name: Hana NovakStudent ID: 007394770**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \approx 0.60$$

redo

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.231) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

$$2 \times .378 = .756$$

Name: Hana NovakStudent ID: 001394770

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = .8 - .1(-.486) = .849$$

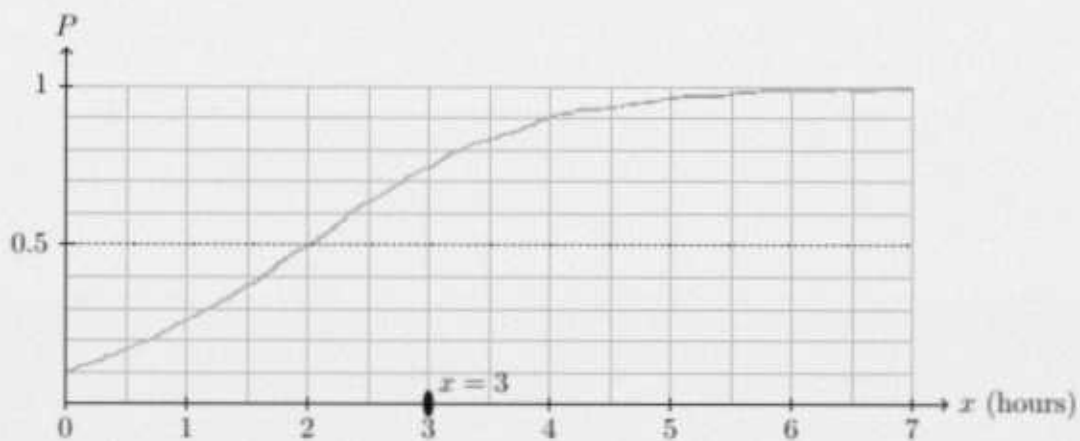
$$b = -.7 - .1(.282) = -.728$$

$$.1029 / .2904$$

$$1 + 1.49 = 2.49$$

x2?

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

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Name: Diego RamosStudent ID: 00445246**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H)P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

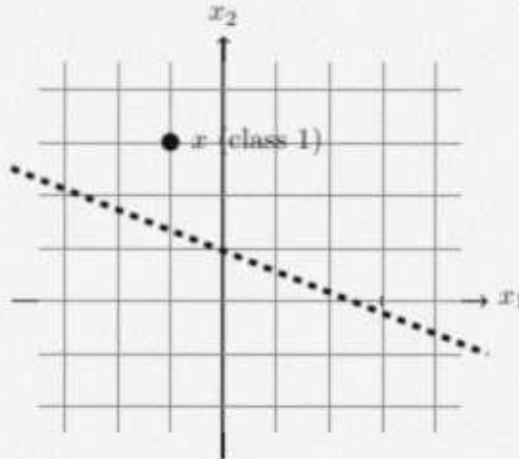
$$= 0.354$$

Name: Diego Ramos

Student ID: 001445246

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$\text{slope} = -3.5 / 9 = -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]^\top: -0.39 + 3 = 0.94$$

(b) not unique: $c \cdot w, c \cdot b, c > 0$

$$w = [0.39, 1], b = -0.94$$

see back →

(cont.)

~~slope~~ $55/8$

$$x_2 = -0.39x_1 + 0.94$$

$$w = [0.39, 1], b = -0.94$$

1029/2904

$$x = [-13]: -39 + 3 - .94$$

Name: Diego RamosStudent ID: 004445216**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

~~$$1/1.251401789$$~~

$$\text{sigmoid}(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) = 0.62$$

$$6 \times 375 = 225$$

$$8/5.5 = 1.4545$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -5, 0, 1, 1.5$$

$$L = -[\ln(.731) + \ln(.672) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= .313 + .475 + .693 + .313 + .201$$

$$\boxed{L = 2.00}$$

Name: Diego RamosStudent ID: 004445246

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

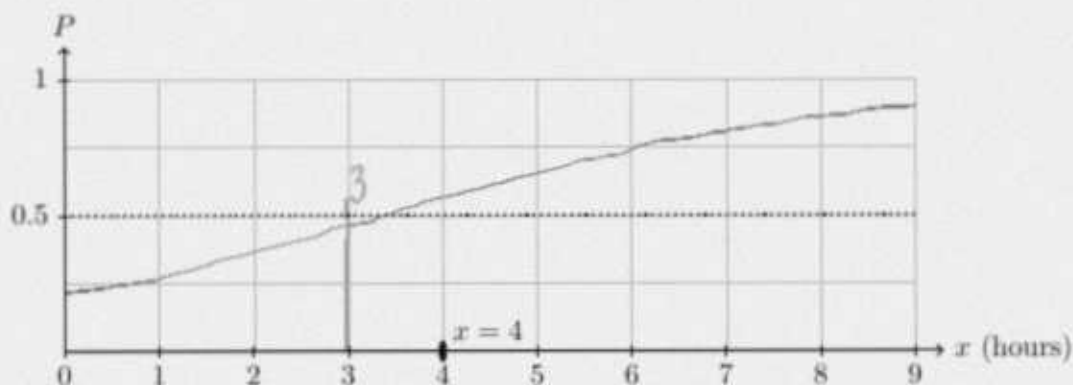
$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -.5 - .05(-.304) = -.485$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Jordan BlakeStudent ID: 006220555**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

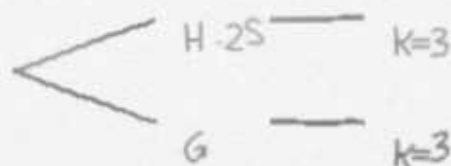
~~$P(H|K=3) = .4116$~~

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .2904$$

~ 0.354



Name: Jordan BlakeStudent ID: 006220555**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

$$35/9 = 389$$

$$2 \times 378 = 756$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$L = 2.31 + .401 + .769$$

$$z = -1, -5, 0, 1, 1.5$$

$$p = .264, .378, .5, .731, .818$$

$$L = -[4 \ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= 3.13 + .475 + .693 + 3.13 + .201$$

$$L \sim 2.00$$

$$.384 \times .4 = .1536$$

check

cont. from front

$$z = -1, -5, 0, 1, 5$$

$$8/55 = 1.4545$$

$$L \sim 200$$

$$e^{-0.4} = .670$$

redo

Name: Jordan BlakeStudent ID: 006220555

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

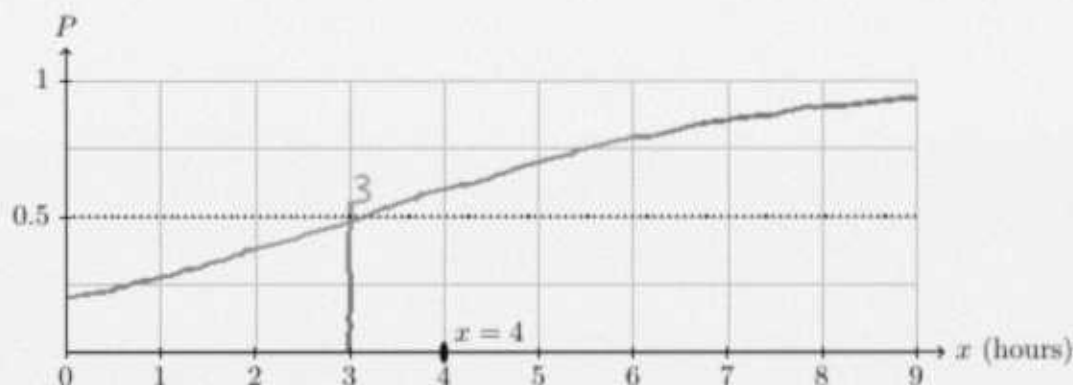
$$dL/db = .269 + .378 - .5 - .269 - .182 = -304$$

$$dL/dw = .282$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -.15 - .05(-304) = -148$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 15 / 5 = 3 \text{ hours}$$

see back →

Q3 cont.

$$.6 \times .375 = .225$$

$$P = .5 \text{ where } z = 0$$

$$w = .5 - .05(-2.91) = .646$$

units?

$$e^{-0.4} = .670$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Name: Ines DugarteStudent ID: 007520139**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

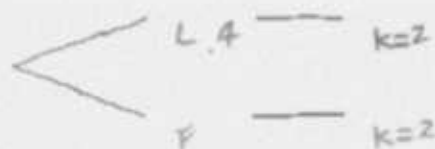
$$P(k=2|L) = .384 \quad P(k=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .1536 + .225$$

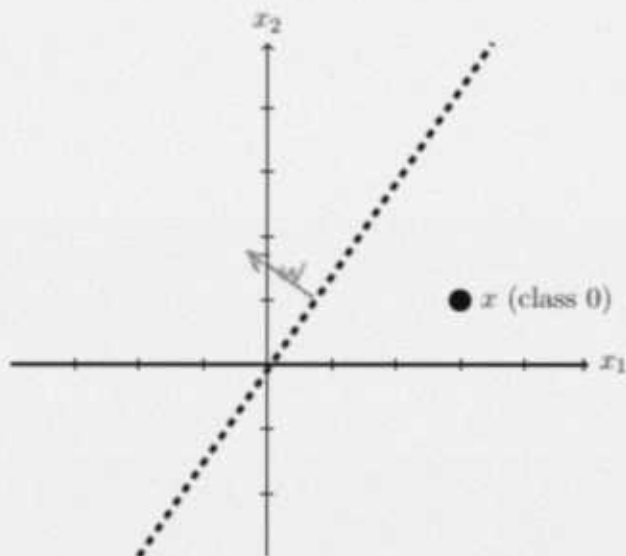
$$\sim 0.406$$

units?



Name: Ines DuarteStudent ID: 007520139**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45 (x_1 + 2)$$

$$\text{slope} = 5.5/8$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

-> class 1, wrong side! flip

(b) no. any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Ives D. ArgeStudent ID: 00752039**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\approx 1 / 1.670$$

$$\text{sigma}(0.4) = 0.4$$

$$P(\text{pass} | x=3) \sim 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

redo

x2?

Name: Ines DuarteStudent ID: 00520139

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

~~$dL/dw = .389$~~

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

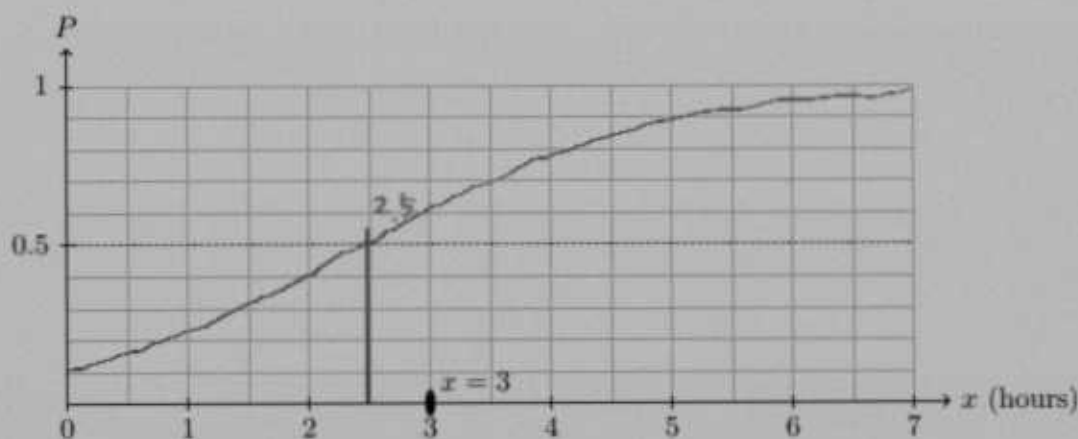
$$dL/db = .231 + .401 - .231 - .119 = .282$$

x2?

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$.6 \times .375 = .225$$

$$3.5 / 9 = .389$$

$$1 + 1.49 = 2.49$$

see back →

cont. from front

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$3.5/9 = .389$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$.384 \times .4 = .1536$$

$$1/2.49 = .401$$

$$\ln .5 = -.693$$

$$.1029/.2904$$

switch where $wx + b = 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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- Calculators are allowed. No notes, no phones.

Name: Taylor BrooksStudent ID: 004920895**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) \approx P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

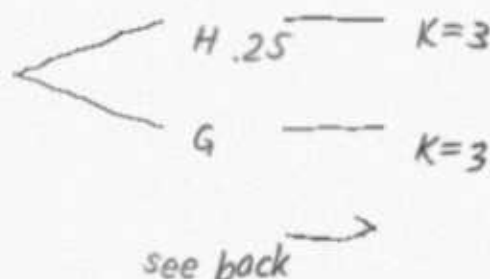
$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$\boxed{\sim 0.354}$$

$$3.5/9 = .389$$



cont. from front

$$P(H|K=3) = .1029 / .2904$$

$$2 \times .378 = .756$$

$$e^{-0.4} = .670$$

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$.6 \times .375 = .225$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$
$$= .2904$$

$$3.5/9 = .389$$

$$\ln .5 = -.693$$

$$1 + 1.49 = 2.49$$

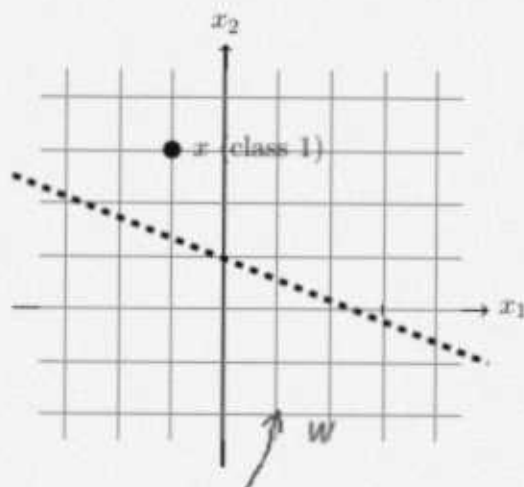
$$1/2.49 = .401$$

Name: Taylor Brooks

Student ID: 004920845

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39 (x_1 + 4)$$

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [9, 3.5]$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$w = [0.39, 1], b = -0.94$$

$$1 + 1.49 = 2.49$$

$$0.6 \times 0.375 = 0.225$$

$$\ln 0.5 = -0.693$$

hmm

Name: Taylor BrooksStudent ID: 004920845**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\approx 1 / 1.607$$

$$P(\text{pass} | x=4) \approx 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$L = .231 + .401 + .769$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

Name: Taylor BrooksStudent ID: 004920845

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

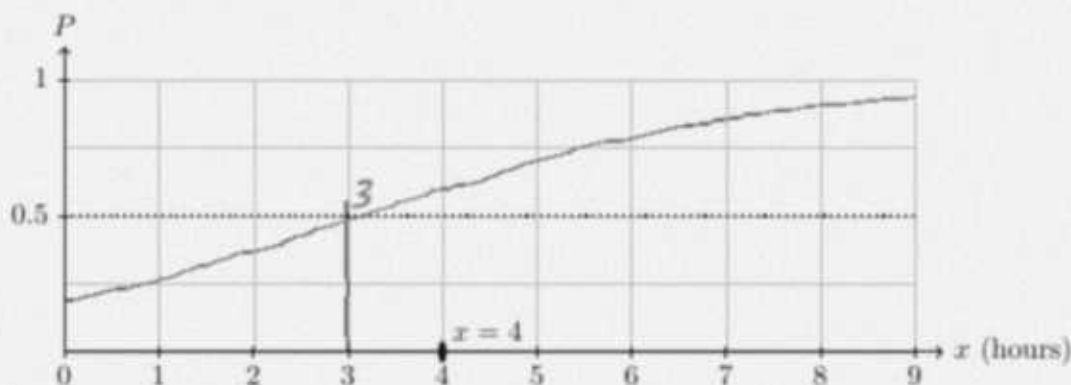
$$\begin{aligned} dL/dw &= .269 + .378(2) - .5(3) - .269(5) - .182(6) \\ &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \end{aligned}$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

!!

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Sofia LindqvistStudent ID: 00916465**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = \frac{P(K=2|L) P(L)}{P(K=2)}$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

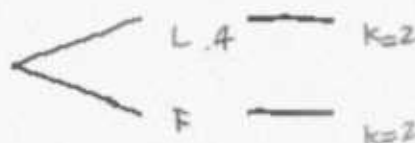
$$= .3786$$

$$\frac{.1536}{.3786} =$$

$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

see back →



Q1 cont.

$$\ln .5 = -.693$$

$$2 \times .378 = .756$$

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(L|K=2) = .1536 / .3786$$

$$= .1536 + .225$$

$$.6 \times .375 = .225$$

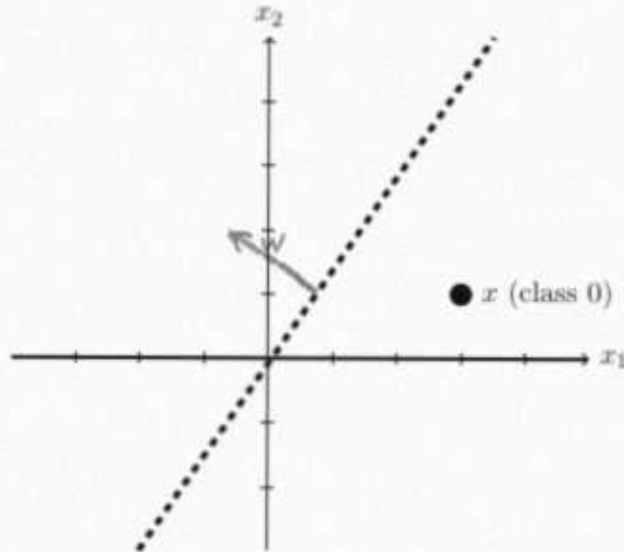
$$= .3786$$

$$= .384(.4) + .375(.6)$$

$$\sim 0.406$$

Name: Sofia LindqvistStudent ID: 009164665**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45 (x_1 + 2)$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$$\text{slope} = 5.5/8$$

-> class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1] \quad b = 0.09$$

see back →

(cont.)

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$$1/2.49 = .401$$

$$w = [-1.45, 1], b = 0.09$$

$$.6 \times .375 = .225$$

Name: Sofia LindqvistStudent ID: 009164665**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\text{Sigmoid}(0.4) = 1 / (1 + e^{-0.4})$$

$$\approx 1 / 1.670$$

$$P(\text{pass} | x=3) \approx 0.60$$

$$e^{-0.4} = .670$$

$$1 + 1.49 = 2.49$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \approx 1.17$$

$$3.5/9 = .389$$

$$.1029 / .2904$$

$$8/5.5 = 1.4545$$

See back 

(cont)

$$\text{sigma}(0.4) = \frac{1}{1 + e^{-0.4}}$$

$$P(\text{PASS} | x=3) \sim 0.60$$

$$e^{-0.4} = .670$$

<

Name: Sofia LindqvistStudent ID: 009164665

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

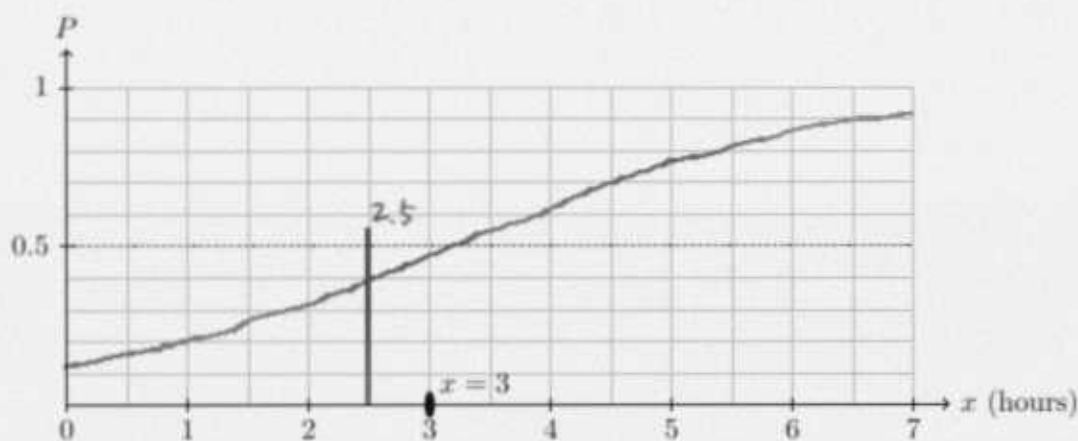
$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



Switch where $w x + b = 0$
 $x = 2 / 0.8 = 2.5 \text{ hours}$

$$1/2.49 = .401$$

$$.384 \times .4 = .1536$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Omar SutherlandStudent ID: 007111858**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = \frac{P(K=3|H) P(H)}{P(K=3)}$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

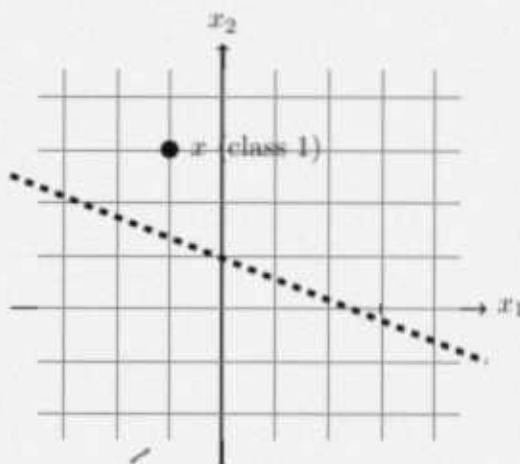
$$\approx 0.354$$

Name: Omar Sutherland

Student ID: 007711858

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39, 1], b = -0.94$$

Name: Omar SutherlandStudent ID: 00711858**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$\begin{aligned} \sigma(0.5) &= 1 / (1 + e^{-0.5}) \\ &= 1 / 1.607 \\ P(\text{pass} | x=4) &\sim 0.62 \end{aligned}$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\begin{aligned} z &= -1, -5, 0, 1, 1.5 \\ p &= .269, .378, .5, .731, .818 \\ L &= -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)] \\ &= .313 + .475 + .693 + .313 + .201 \\ L &\sim 2.00 \\ 3.5/9 &= .389 \end{aligned}$$

Name: Omar Sutherland

Student ID: 007711858

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

~~$dL/dw = .282$~~

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

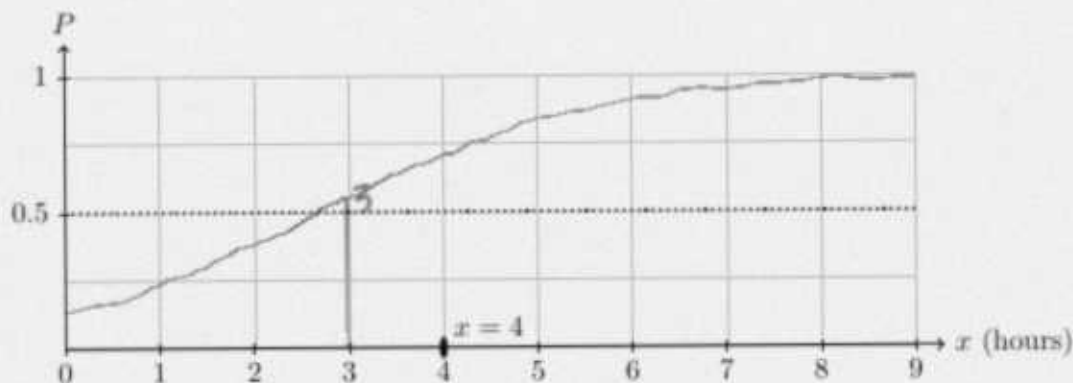
$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .6455$$

$$b = .15 - .05(-.304) = .1652$$

- (d) Sketch $P(\text{pass} | x)$ under the original (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$P = .5$ where $z = 0$
 $x = 1.5 / 5 = 3$ hours

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Name: Jonah PierceStudent ID: 002595843**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$= .4116(.25) + .25(.75)$$

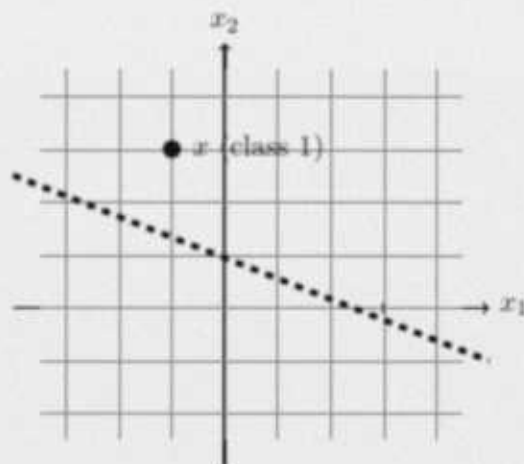
$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

Name: Jonathan PierceStudent ID: 002595843**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{K}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x = [-1, 3]: -0.39 + 3 = .99$$

$$\overline{b = 3}$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = 0.99$$

$$1/2 \cdot 49 = .401$$

(cont.)

$$x_2 = -0.39x_1 + 0.94$$

$$.384 \times .4 = .1536$$

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$8/5.5 = 1.4545$$

$$w = [0.39, 1], b = -0.94$$

$$e^{-0.4} = .670$$

(b) not unique: $cw, cb, c > 0$

Name: Jonah PierceStudent ID: 002595843**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\text{sigmoid}(0.5) = 1 / (1 + e^{-0.5})$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + [\ln(.5) + \ln(.731) + \ln(.818)]]$$

$$L \sim 2.00$$

Name: Jonah PierceStudent ID: 002595843**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

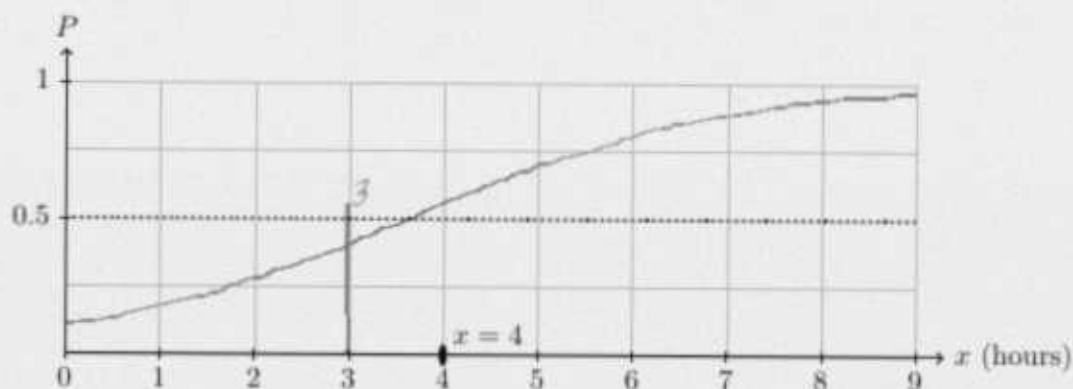
$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

?

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Theo CastilloStudent ID: 002245104**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$
$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

see back $\rightarrow$

(cont.)

$$8/55 = 1.4545$$

$$3.5/9 = .389$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

ans

$$P(H|K=3) = .4116$$

$$= .1029 + .1875$$

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$384 \times .4 = 1536$$

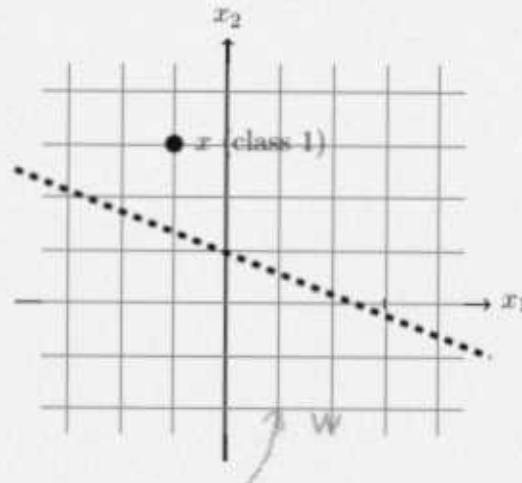
$$\ln 5 = -693$$

Name: Theo Castillo

Student ID: 002295104

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}(w^T x + b > 0)$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\begin{aligned} \text{slope} &= (-1 - 2.5) / (5 - (-4)) \\ &= -3.5 / 9 \sim -0.39 \\ x_2 - 2.5 &= -0.39 (x_1 + 4) \\ x = [-1, 3]: &-0.39 + 3 = .94 \\ &= 1.67 > 0 \rightarrow \text{class 1 ok} \end{aligned}$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39 \ 1] \quad b = -0.94$$

Name: Theo CastilloStudent ID: 002295104**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$p = e^z$$

$$z = 0.5(4) - 1.5 = 0.5$$

$$\text{sigma}(0.5) = 1 / (1 + e^{-0.5})$$

$$\approx 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

hmm

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= .313 + .475 + .693 + .313 + .201$$

$$\text{sigma}(0.4) = 0.4$$

$$L \sim 2.00$$

$$e^{-0.4} = .670$$

$$1/2.49 = .401$$

Q3 cont.

$$384 \times .4 = 1536$$

$$\text{sigma}(0.5) = 1 / (1 + e^{-0.5})$$

$$\ln .5 = -.693$$

$$L = -[\ln(731) + \ln(622) + \ln(5) + \ln(731) + \ln(818)]$$

~~ln(0)??~~

$$2 \times .378 = .756$$

$$1/249 = .401$$

$$z = 0.5(4) - 1.5 = 0.5$$

$$P(\text{pass} | x=4) \sim 0.62$$

$$6 \times 375 = 225$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points

Name: Lena HoffmannStudent ID: 002920174**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

$$= .3786$$

$$\textcircled{-.406}$$

$$e^{-0.4} = .670$$

$$.6 \times .375 = .225$$

see back →

Q1 cont.

$$= .384(.4) + .375(.6)$$

$$1 + 1.49 = \overline{2.49}$$

$$P(L|K=2) = .384$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$= 0.406$$

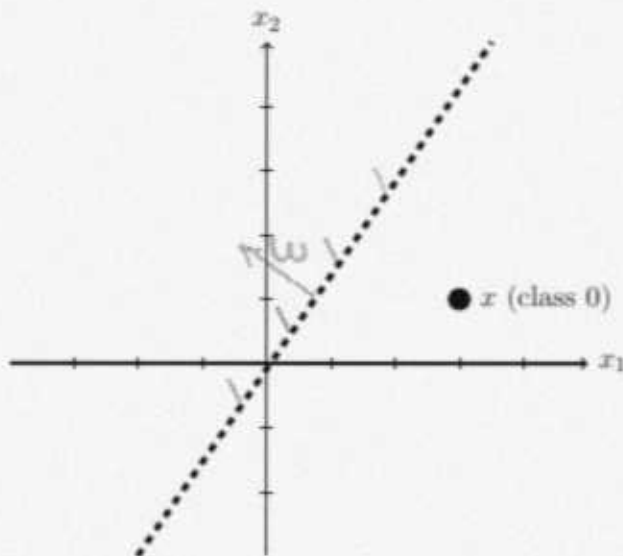
$$.6 \times .375 = .225$$

$$8/5.5 = 1.4545$$

$$P(L|K=2) = .1536 / .3786$$

Name: Lena HoffmannStudent ID: 002920131**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \quad \text{See 2}$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

→ class 1, wrong side! flip

$$w \cdot x + b < 0?$$

$$w = [-1.45, 1], b = 0.09$$

see back →

Q2 cont.

$$1 + 1.49 = 2.49$$

$$\ln .5 = -.693$$

$$x_2 + 3 = 1.45 (x_1 + 2)$$

$$1/2.49 = .401$$

Name: Lena HoffmannStudent ID: 002920171**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

/info??

$$P(\text{pass} | x=3) = 0.60$$

$$\ln .5 = -.693$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

Name: Lena HoffmannStudent ID: 002920171

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .924(4) - .595(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

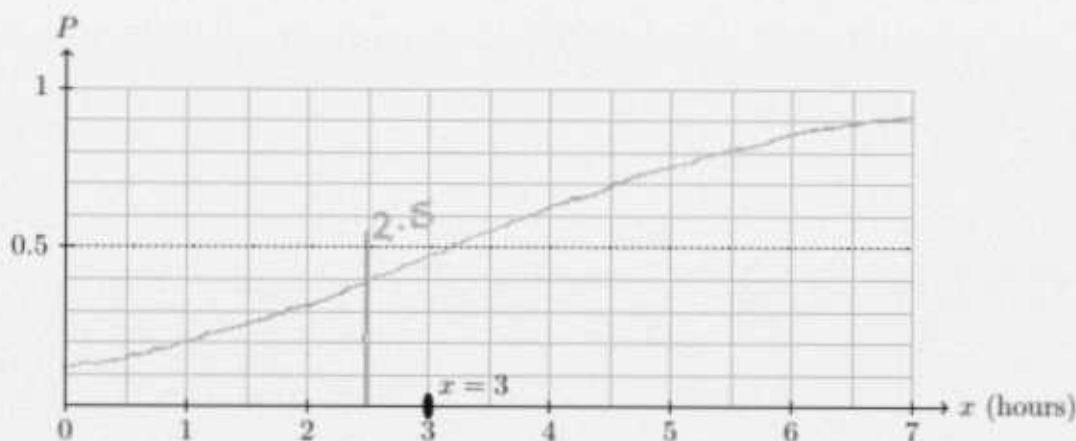
$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

$$\ln .5 = -.693$$

$$e^{-0.4} = .670$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$



Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Name: Kai YamamotoStudent ID: 007138488**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(L \mid K=2) &= P(K=2 \mid L) P(L) / P(K=2) \\
 P(K=2 \mid L) &= .384 \\
 P(K=2) &= P(K=2 \mid L) P(L) + P(K=2 \mid F) P(F) \\
 &= .384(.4) + .375(.6) \\
 &= .3786 \\
 P(L \mid K=2) &= .1536 / .3786 \\
 &\sim .406 \\
 &.1029 / .2904
 \end{aligned}$$

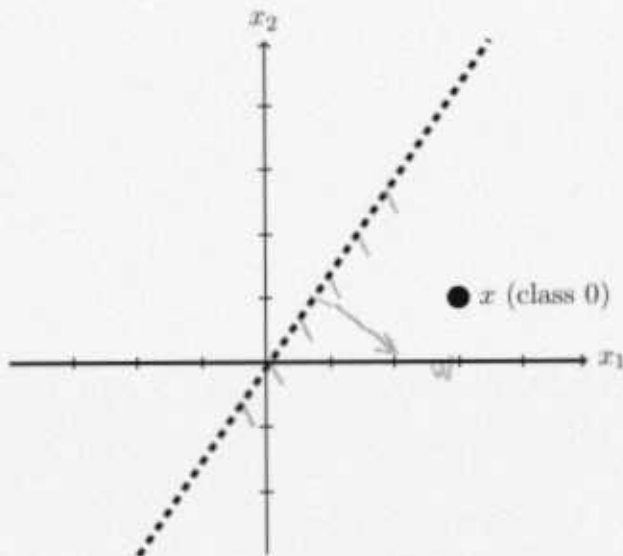
Diagram illustrating the calculation of $P(K=2)$:

```

graph LR
    L[L] -- .4 --> K2[K=2]
    F[F] -- .6 --> K2
    K2 --> P2[.3786]
  
```


Name: Kai YamamotoStudent ID: 007138988**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$$w \cdot x + b > 0?$$

(b) no, any $c > 0$ scales (w, b)

$$w = [1.45, 1], b = 0.09$$

see back →

Q2 cont.

$$\ln .5 = \frac{1.03}{1.45}$$

$$w = [1, 1.45]$$

→ class 1, wrong side! flip

$$.6 \times .375 = .225$$

$$w = [1.45, -1], b = -0.09$$

$$xz = 1.45x_1 - 0.09$$

(b) no, any $c > 0$ scales (w, b)

$$.1029 / .2904$$

$$z \times .378 = .756$$

Name: Kai YamamotoStudent ID: 007138488**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\begin{aligned} \sigma(0.4) &= 0.4 \\ z &= 0.8(3) - 2 = 0.4 \\ \sigma(0.4) &= 1 / (1 + e^{-0.4}) \\ &= 1 / 1.670 \\ P(\text{pass} | x=3) &\sim 0.60 \\ 8/5.5 &= 1.4545 \end{aligned}$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\begin{aligned} \ln(0) &?? \\ z &= -1.2, -0.4, 1.2, 2 \\ p &= .231, .401, .769, .881 \\ L &= -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)] \\ &= .263 + .513 + .263 + .127 \\ L &\sim 1.17 \end{aligned}$$

$$\begin{aligned} 8/5.5 &= 1.4545 \\ .1029/.2904 \end{aligned}$$

Name: Kai YamamotoStudent ID: 007138488

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

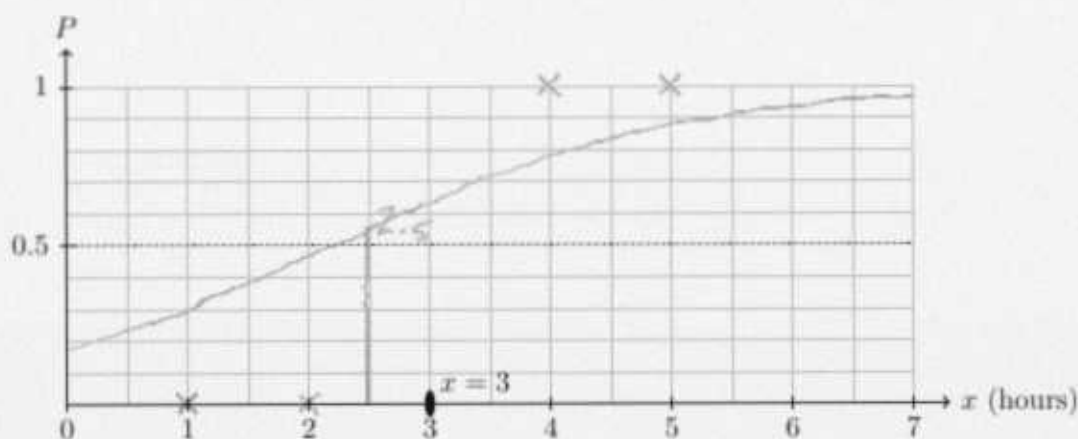
$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = 8 - 1(-.486) = 8.49$$

$$b = 2 - 1(.282) = 1.718$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$
 $x = 2 / 0.8 = 2.5$ hours

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points

Name: Mina HaddadStudent ID: 009597897**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

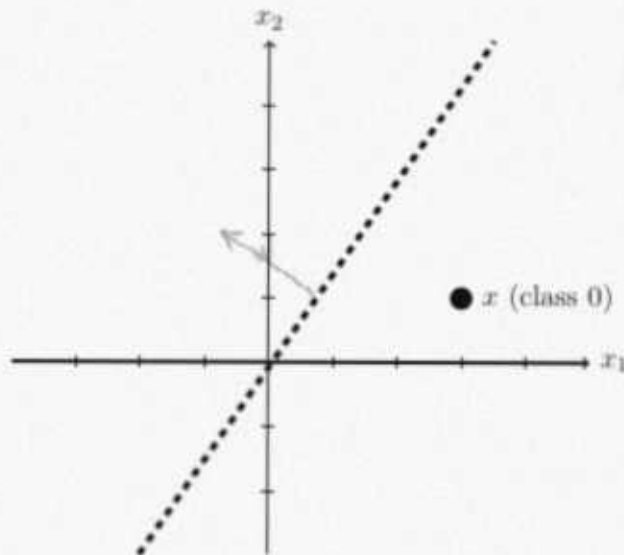
k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(L \mid K=2) &= \frac{P(K=2 \mid L)P(L)}{P(K=2 \mid L)P(L) + P(K=2 \mid F)P(F)} \\
 P(K=2 \mid L) &= 0.384 \\
 P(K=2 \mid F) &= 0.375 \\
 P(K=2) &= P(K=2 \mid L)P(L) + P(K=2 \mid F)P(F) \\
 &= 0.384(0.4) + 0.375(0.6) \\
 &= 0.1536 + 0.225 \\
 &= 0.3786 \\
 &= \frac{0.384(0.8)^2(0.2) + 0.375(0.2)^2(0.8)}{0.384(0.8)^2(0.2) + 0.375(0.2)^2(0.8)} \\
 &\sim 0.406
 \end{aligned}$$

Name: Mina HaddadStudent ID: 009597897**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$x = [3, 1]: 4.36 - 1 = .09 > 0$$

→ class 1, wrong side! flip

~~Let $w^\top x + b > 0$ scales (w, b)~~

$$\text{slope} = 5.5/8$$

$$w = [-1.45, 1], b = 0.09$$

Name: Mina HaddadStudent ID: 009597897**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\sigma(0.4) = \frac{1}{1 + e^{-0.4}}$$

$$P(\text{pass} | x=3) \sim 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

see back $\rightarrow$

units?

Q3 cont.

$$8/5.5 = 1.4545$$

$$1/1.5 = .693$$

$$.6 \times .375 = .225$$

$$.384 \times .4 = .1536$$

$$2 \times .378 = .756$$

$$Z = 0.8(3) \quad Z = 0.4$$

Name: Mina HaddadStudent ID: 009577897**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

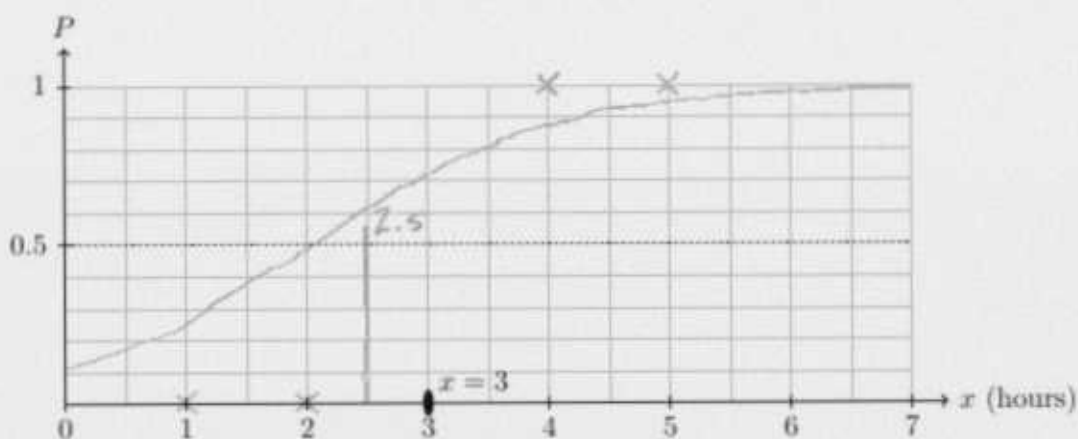
$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486 \rightarrow \text{hmm}$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$
 $x = 2 / 0.8 = 2.5 \text{ hours}$

check
 ☆

Q3 cont.

$$w = .8 - .1(-.486) = .849$$

$$.6 \times .375 = .225$$

$$1/2.49 = .401$$

$$b = -2 - .1(.282) = -2.03$$

$$.1029/.2904$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Name: Avery ChenStudent ID: 008280321**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$= .384(.4) + .375(.6)$$

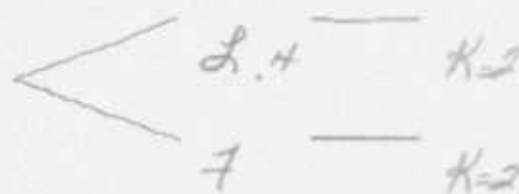
$$= .1536 + .225$$

$$= .3786$$

$$P(L|K=2) = .1536 / .3786$$

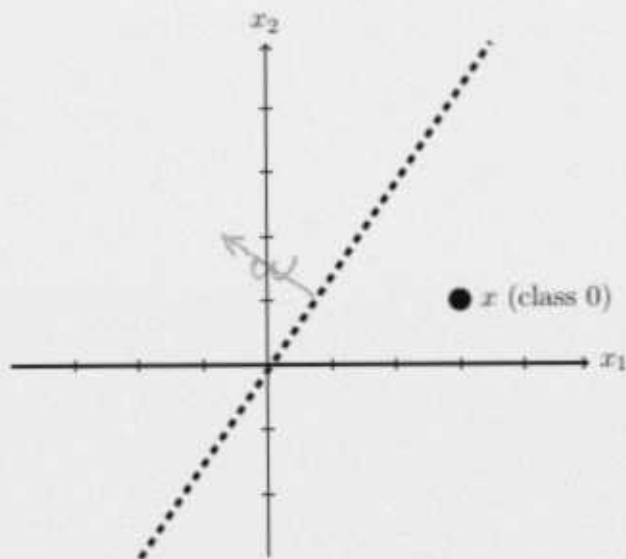
$$0.406$$

ans



Name: Avery ChenStudent ID: 008280527**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45x_1 - 0.09$$

$$x = [3, 1]: 4.36 - 1 - 0.09 > 0$$

→ class 1, wrong side! flip
(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Avery ChenStudent ID: 008280327**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\bar{z} = 0.8(3) - 2 = 0.4$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

$$1 + 1.49 = 2.49$$

$$1/2.49 = .401$$

$$3.5/9 = .389$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\bar{z} = -1.2, -0.4, 1.2, 2$$

 $\ln(0)$??

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

$$1 + 1.49 = 2.49$$

$$8/5.5 = 1.4545$$

Name: Avery ChenStudent ID: 008280527

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

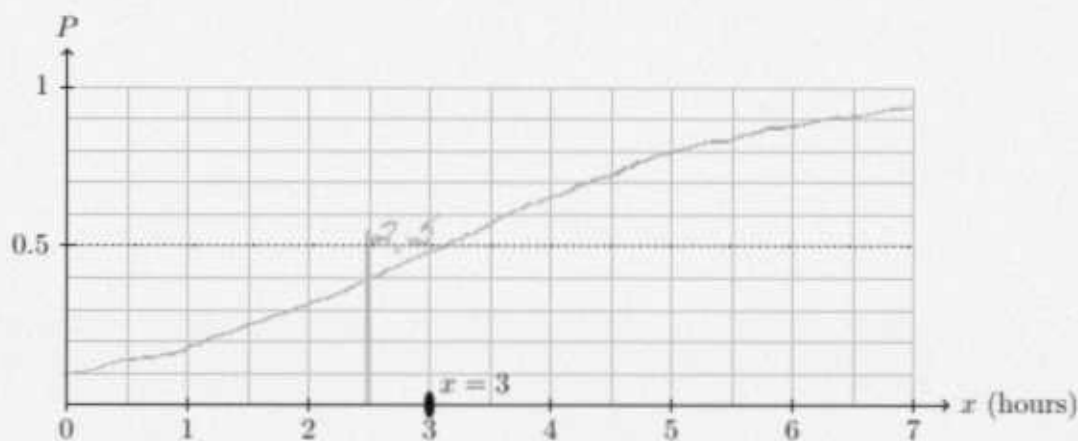
$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$\boxed{w = .8 - .1(-.486) = .849}$$

$$\boxed{b = 2 - .1(.282) = 1.972}$$

12?

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points

Name: Marcus WebbStudent ID: 007911523**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= 4116(.25) + 25(.75)$$

$$= .384(.6) + .375(.4)$$

$$= .1029 + .1875$$

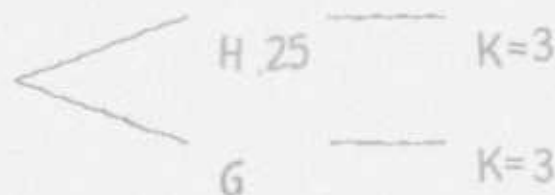
$$= .2904$$

$$\sim 0.354$$

$$3.5/9 = .389$$

$$1 + 1.49 = 2.49$$

$$1/2.49 = .401$$

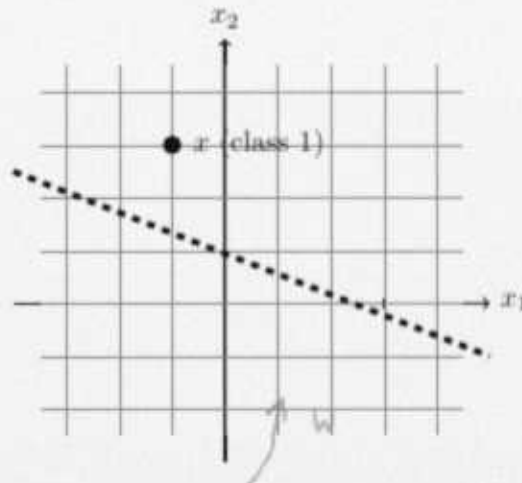


Name: Marcus Webb

Student ID: 007977523

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^T x + b > 0\}$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 \sim .94$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$.1029 / .2904$$

$$2 \times .378 = .756$$

$$.384 \times .4 = .1536$$

Name: _____

Student ID: _____

Marcus Webb

007977523

Problem 3 [10 pts] Pass or fail

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\text{sigma}_d(0.5) = 1 / (1 + e^{-0.5})$$

$$\ln(0) ??$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

Name: Marcus WebbStudent ID: 007977523**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\begin{aligned} dL/dw &= .269 + .378(2) - .5(3) - .269(5) - .182(6) \\ &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \end{aligned}$$

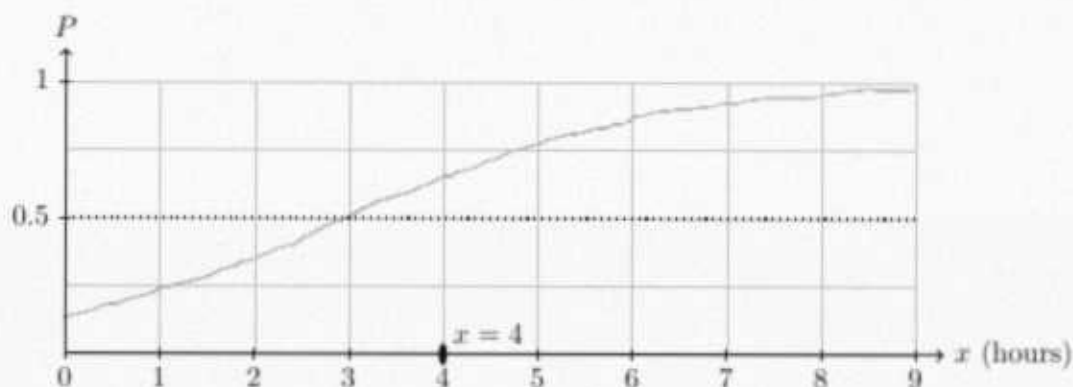
$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

units?

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$\begin{aligned} P &= .5 \text{ where } z = 0 \\ x &= 1.5 / .5 = 3 \text{ hours} \end{aligned}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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- Calculators are allowed. No notes, no phones.

Name: Priya NatarajanStudent ID: 008453099**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

$$= .3786$$

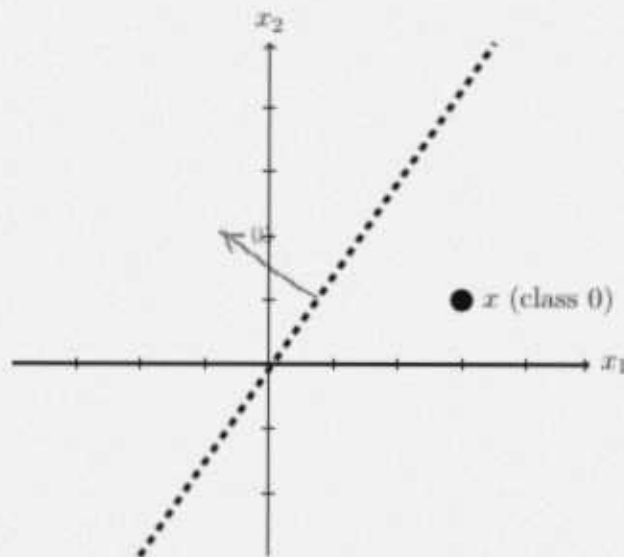
$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

units?

Name: Priya NatarajanStudent ID: 008453099**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$w = [1, 1.45]$$

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \approx 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

→ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Divya NatarajanStudent ID: 008453099**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\text{sigma}(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

Name: Priya NatarajanStudent ID: 008453099**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

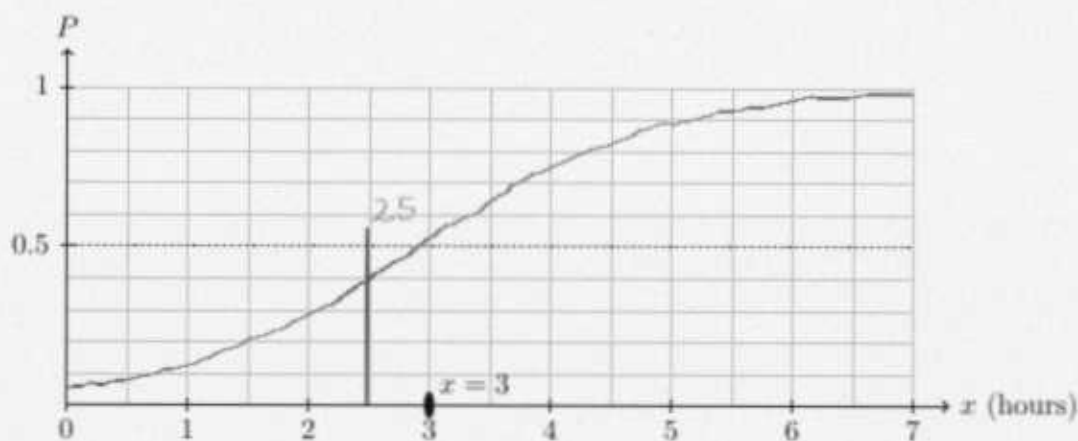
$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$b = 2 - .1(.282) = 2.02$$

$$1 + 1.49 = 2.49$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Riley MorenoStudent ID: 00222695**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

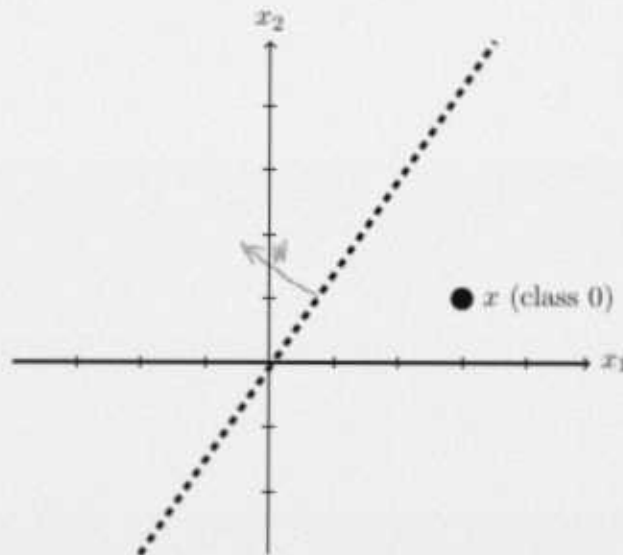
$$\begin{aligned}
 P(L|K=2) &= P(K=2|L)P(L) / P(K=2) \\
 P(K=2|L) &= .384 \quad P(K=2|F) = .375 \\
 P(K=2) &= P(K=2|L)P(L) + P(K=2|F)P(F) \\
 &= .384(.4) + .375(.6) \\
 &= .1536 + .225 \\
 &= .3786 \\
 P(L|K=2) &= .1536 / .3786 \\
 &= .406 \\
 \ln .5 &= -.693
 \end{aligned}$$

Name: Riley Moreno

Student ID: 00222695

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 = 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - 0.09 > 0$$

→ class 1, wrong side! Flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

$$1/2.49 = 0.4$$

$$1 + 1.49 = 2.49$$

Name: Ridg MorenoStudent ID: 002122695**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} \mid x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} \mid x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\text{sigmoid}(0.4) = 1 / (1 + e^{-0.4})$$

$$P(\text{pass} \mid x=3) = 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$L = -[\ln(0.69) + \ln(0.99) + \ln(0.769) + \ln(0.880)]$$

$$= 263 + 513 + 263 + 127$$

$$L = 1.17$$

$$8/55 = 14545$$

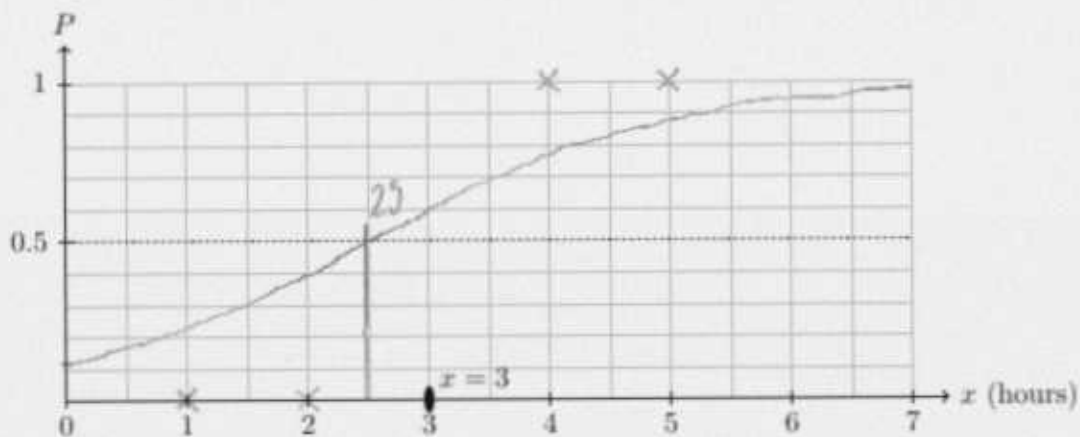
Name: Riley MorenoStudent ID: 002022695

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$\begin{aligned} dL/dw &= 231(1) + 401(2) - 231(4) - 119(5) \\ &= 231 + 802 - 924 - 595 = -486 \\ dL/db &= 231 + 401 - 231 - 119 = 282 \\ w &= 8 - 1(-486) = 849 \\ b &= -2 - 1(282) = -284 \end{aligned}$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$
 $x = 2 / 0.8 = 2.5$ hours

redo

cont. from front

$$1/249 = .401$$

$$6 \times .375 = .225$$

$$\Delta w = .8 + .1(-486)$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$2 \times .378 = .756$$

$$3.5/9 = .389$$

$$1029/2904$$

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5)$$

$$1 + 1.49 = 2.49$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Name: Felix AdeyemiStudent ID: 004285862**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

Q1 cont.

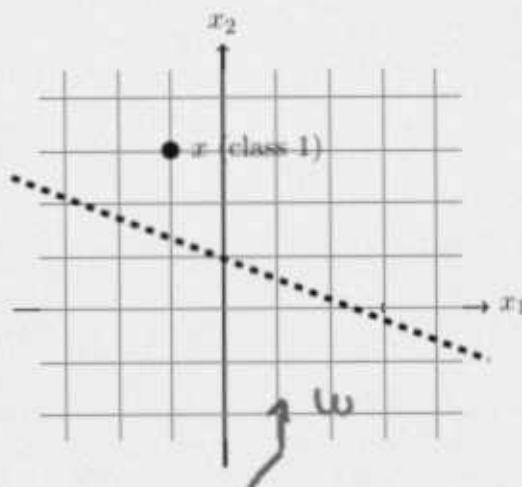
$$P(H|K=3) = \frac{P(K=3|H) P(H)}{P(K=3)}$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$1 + 1.49 = 2.49$$

Name: Felix AdeyemiStudent ID: 004285862**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= -3.5 / 9 = -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$= 2.61 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

Name: Felix AdeyemiStudent ID: 004285862**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} \mid x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} \mid x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

x2?

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} \mid x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

Name: Felix AdeyemiStudent ID: 004285862

Problem 3 (continued)

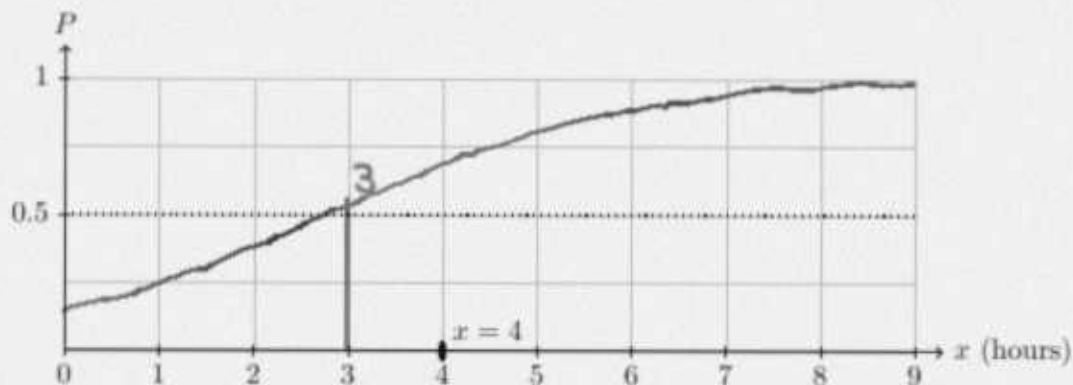
- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\begin{aligned} dL/dw &= .269 + .378(2) - .5(3) - .269(5) - .182(6) \\ &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \end{aligned}$$

$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$p = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

see back →

cont. from front

$$w = .5 - .05(-2.91) = .646$$

$$\cancel{x = 0.6}$$

$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$\ln .5 = -.693$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Grace KimStudent ID: 006071844**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = \frac{P(K=2|L)P(L)}{P(K=2|L)P(L) + P(K=2|F)P(F)}$$

$$P(K=2|L) = .384$$

$$P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

$$= .3786$$

units?

$$P(L|K=2) = .1536 / .3786$$

$$.384 \times .4 = .1536$$

see back

(cont.)

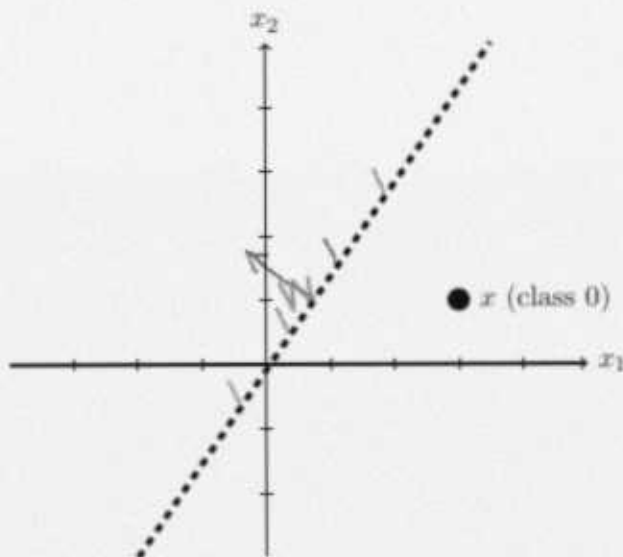
$$1/2.49 = .401$$

$$8/5.5 = 1.4545$$

$$P(L|K=2) = .1536 / .3786$$

Name: Grace KimStudent ID: 006071844**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2)) = 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1] \quad b = -0.09$$

$$x = [3.5, 1] \quad 3.6 - 1 - .09 > 0$$

$$w = [1, 1.45]$$

$$(b) \text{ via any } c > 0 \text{ scaled } (w, b)$$

$$w = [-1.45, 1], \quad b = 0.09$$

$$1029/2904$$

Name: _____

Student ID: _____

Grace Kim

00607844

Problem 3 [10 pts] Pass or fail

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\begin{aligned} z &= 0.8(3) - 2 = 0.4 \\ \sigma(0.4) &= 1 / (1 + e^{-0.4}) \\ &= 1 / 1.670 \\ P(\text{pass} | x=3) &\sim 0.60 \\ \ln 5 &= -0.693 \\ 3.84 \times .4 &= .1536 \end{aligned}$$

redo

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\begin{aligned} z &= -1.2, -0.4, 1.2, 2 \\ L &= -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)] \\ L &\sim .263 + .513 + .263 + .127 \\ L &\sim 1.17 \end{aligned}$$



Name: _____

Student ID: _____

Grace Kim

006071844

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .23(1) + .40(2) - .23(4) - .19(5)$$

$$= .231 + .802 - .924 - .955 = -.846$$

$$dL/db = .231 + .401 - .231 - .19 = .282$$

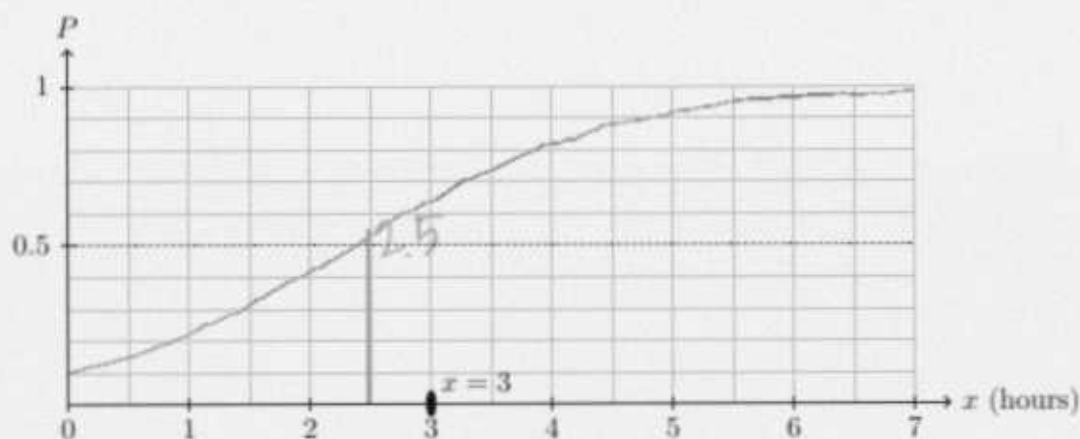
$$w = .8 - .1(-.846) = .849$$

$$b = -2 - .1(.282) = -2.03$$

$$3.5/9 = .389$$

$$e^{-0.4} = .670$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$x = 2 / 0.8 = 2.5 \text{ hours}$$

see back →

cont. from front

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$6 \times 375 = 225$$

$$dL/dW = .23(1) + .40(2) - .23(4) - .19(5)$$

~~$$284 \times 4 = 1136$$~~

$$W = .8 + .1(-486)$$

$$1 + 1.49 = 2.49$$

x2?

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Name: _____

Sam Okafor

Student ID: _____

009030878

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$\sim 0.354$$

$$.6 \times .375 = .225$$

see back 

Q1 cont.

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$\sim 0.354$$

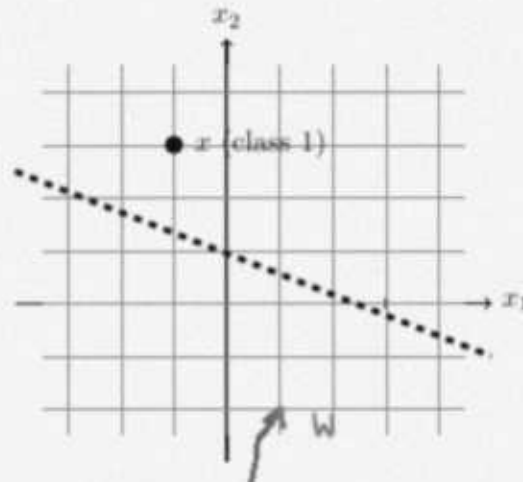
$$= .1029 + .1875$$

Name: Sam Okafor

Student ID: 009030878

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x = [-1, 3]: -0.39 + 3 = .94$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$.1029 / .2904$$

$$.384 \times .4 = .1536$$

units?

redo

see back →

(cont.)

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

$$x_2 = -0.39x_1 + 0.94$$

$$3.5/9 = .389$$

$$2 \times .378 = .756$$

$$= -3.5/9 \sim -0.39$$

$$\ln .5 = -.693$$

$$w = [0.39, 1], b = -0.94$$

(b) not unique: $cw, cb, c > 0$

$$1 + 1.49 = 2.49$$

$$8/5.5 = 1.4545$$

Name: Sam OkaforStudent ID: 009030878**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

~~$$z = 0.5(4) - 1.5 = 0.5$$~~

$$L = .231 + .401 + .769$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$\sigma(0.4) = 0.4$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

Name: _____

Student ID: _____

Sam Okafor

009030878

Problem 3 (continued)

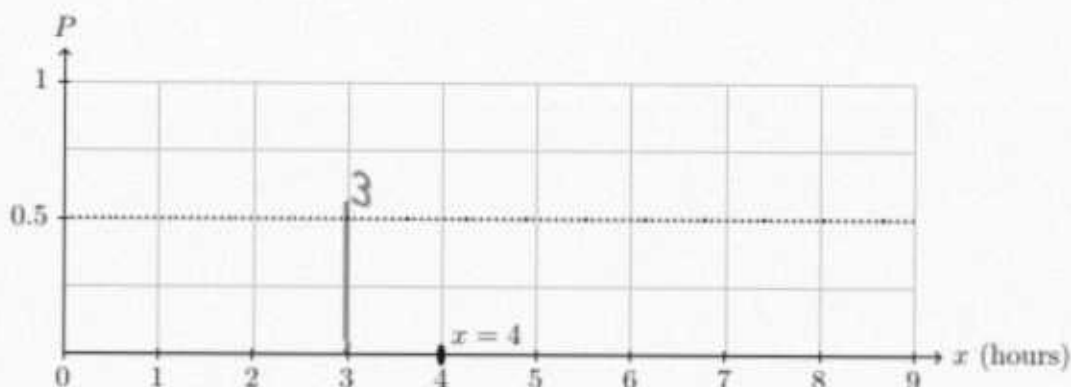
- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.94) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / 5 = 3 \text{ hours}$$

$$.1029 / .2904$$

$$8 / 5.5 = 1.4545$$

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Probability and linear classifiers
(fictional course, for the scan_split example)

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Name: _____

Student ID: _____

Noor Rahman

008087777

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(H|K=3) &= P(K=3|H) P(H) / P(K=3) \\
 P(K=3|H) &= .4116 \quad P(K=3|G) = .25 \\
 P(K=3) &= P(K=3|H)P(H) + P(K=3|G)P(G) \\
 &= .4116(.25) + .25(.75) \\
 &= .1029 + .1875 \\
 &= .2904
 \end{aligned}$$

$$\boxed{P(H|K=3) = .1029 / .2904} \\
 \sim 0.354$$

$$\begin{aligned}
 .384 \times 4 &= 1.536 \\
 2 \times .378 &= .756
 \end{aligned}$$

see back $\rightarrow$

(cont.)

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .2904$$

$$8/5.5 = 1.4545$$

$$= .1029 + .1875$$

$$P(K=3|H) = .4116$$

$$P(K=3|G) = .25$$

$$.384 \times .4 = .1536$$

$$\sim 0.354$$

$$1 + 1.49 = 2.49$$

units?

Name: _____

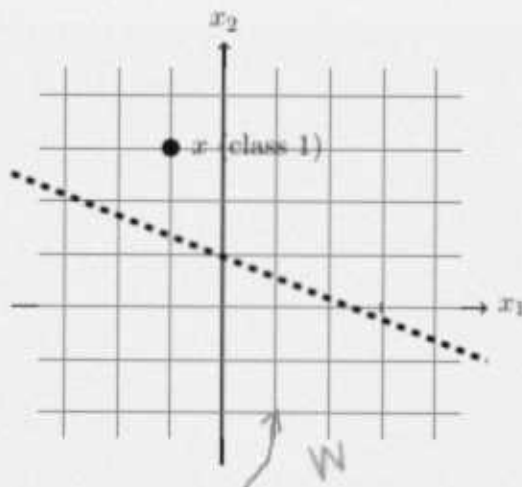
Student ID: _____

Noor Rahman

008087777

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mu[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = \frac{(-1 - 2.5)}{(5 - (-4))} = -3.5 / 9 \sim -0.39$$

$$w \cdot x + b < 0 ?$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39 (x_1 + 4)$$

$$x_2 = -0.39 x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

$$w = [0.39, 1] \quad b = -0.94$$

$$8/5.5 = 1.4545$$

Name: Noor RahmanStudent ID: 008087777**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\text{sigm}(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$L = 2.31 + 4.01 + 7.69$$

$$p = 0.269, 0.378, 0.5, 0.731, 0.88$$

$$L = -[\ln(0.731) + \ln(0.622) + \ln(0.5) + \ln(0.731) + \ln(0.88)] \times 2?$$

$$L \approx 2.00$$

$$8/5.5 = 1.4545$$

$$2 \times 0.378 = 0.756$$

Name: Noor RahmanStudent ID: 008081777**Problem 3 (continued)**

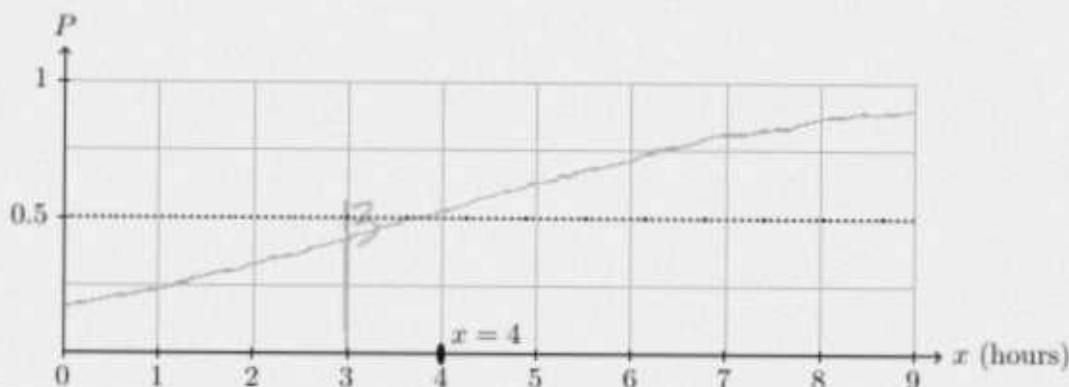
- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\begin{aligned} dL/dw &= .269 + .378(2) - .5(3) - .269(5) - .182(6) \\ &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \end{aligned}$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -3.04$$

$$w = .5 - .05(-2.91) = .646$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$x = 1.5 / .5 = 3 \text{ hours}$$

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Probability and linear classifiers
(fictional course, for the scan_split example)

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- Calculators are allowed. No notes, no phones.

Name: Leo FischerStudent ID: 002982608**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = \frac{P(K=3|H) P(H)}{P(K=3)}$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$= 0.354$$

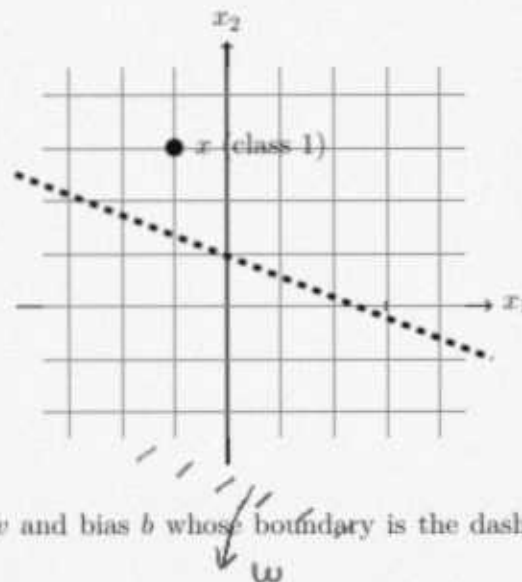
redo

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Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= -3.5 / 9 \approx -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$\approx 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$1029 / 2904$$

$$8 / 5.5 = 1.4545$$

$$e^{-0.4} = 0.670$$

q2 cont.

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39 (x_1 + 4)$$

$$.6 \times .375 = .225$$

$$1 + 1.49 = 2.49$$

(b) not unique: $c_w, c_b, c > 0$

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

Name: Leo FischerStudent ID: 002982605**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$p(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

units?

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Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

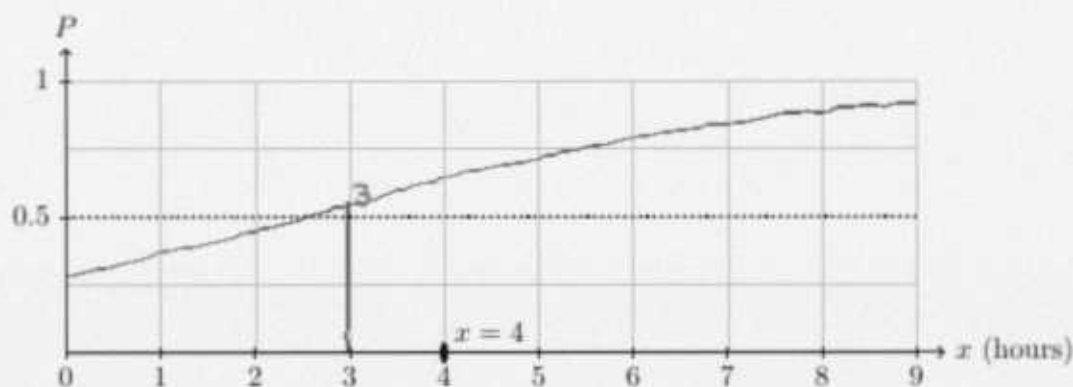
$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .646$$

$$b = 1.5 - .05(-.304) = 1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

